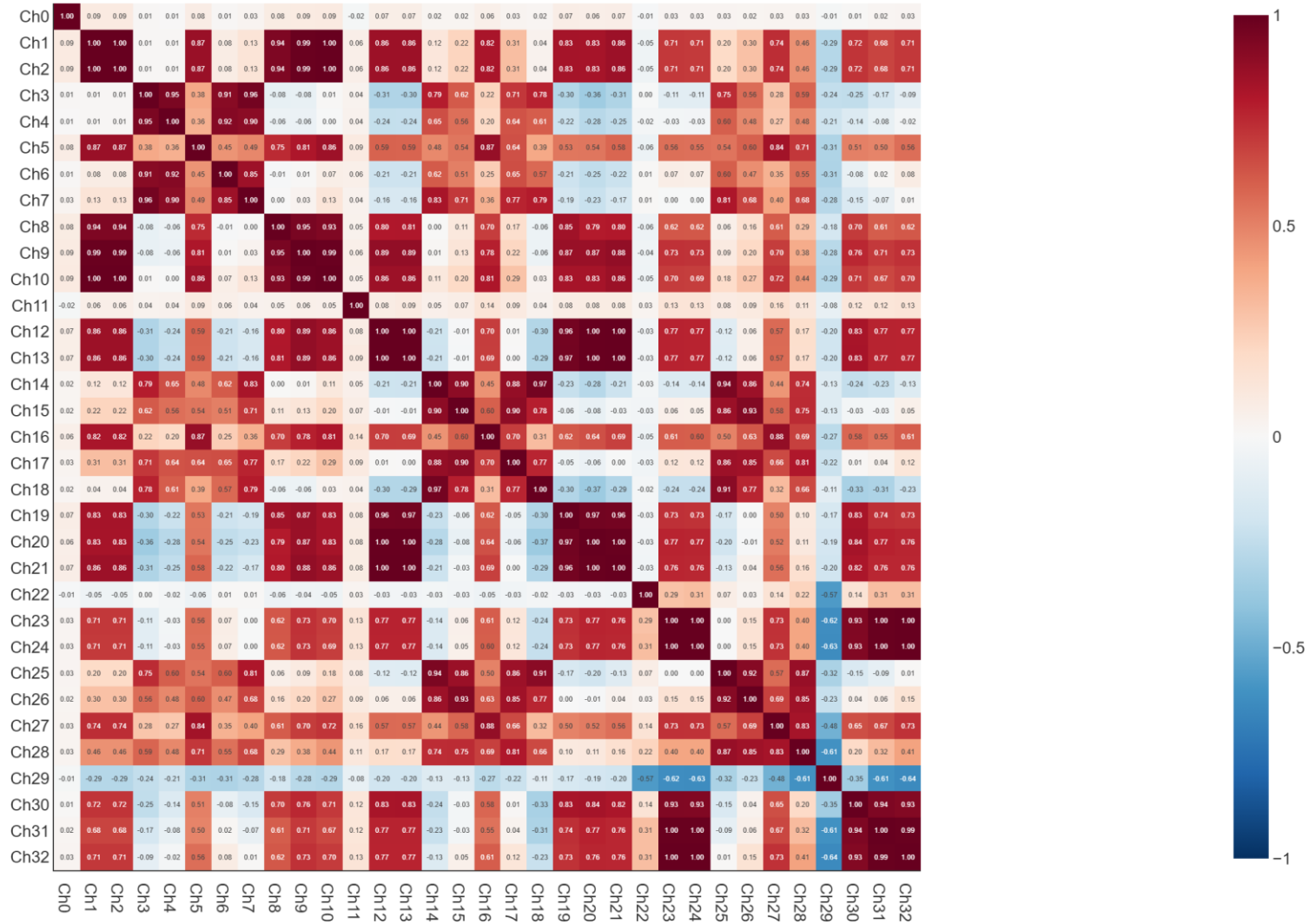
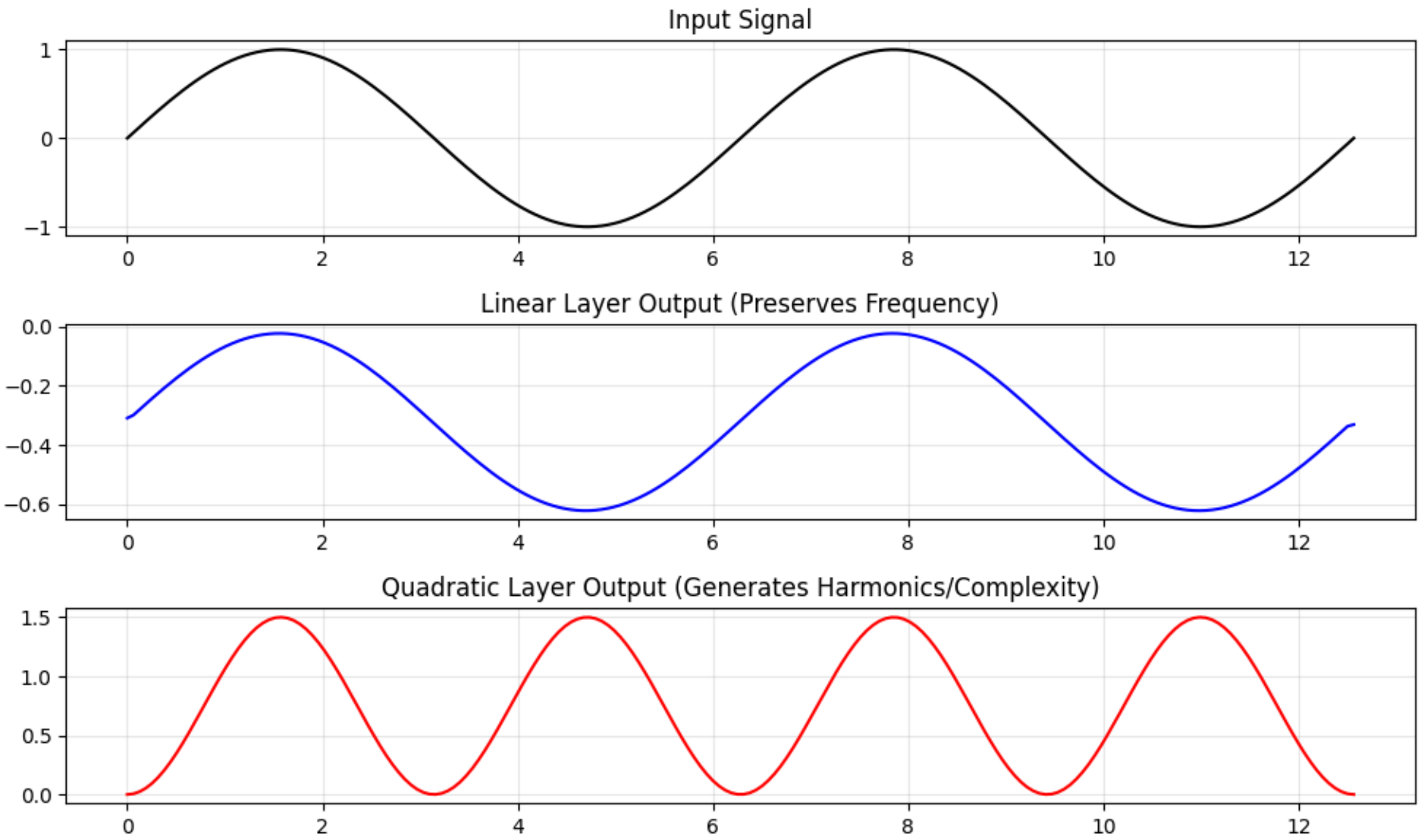
Feature/Channel Correlation Matrix




PCA Projection (Explained Variance 82.2%)
Class
Normal
Inner_014
Ball_014
Ball_007
Ball_021
Outer_007
Inner_007
Inner_021
Outer_014
Outer_021
PC2
PC1
15
10
5
0
−5
−5
0
5
10
15


Input Signal
1
0
−1
0
2
4
6
8
10
12
Linear Layer Output (Preserves Frequency)
0.0
−0.2
−0.4
−0.6
0
2
4
6
8
10
12
Quadratic Layer Output (Generates Harmonics/Complexity)
1.5
1.0
0.5
0.0
0
2
4
6
8
10
12